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II.12

Interspecific affinities among guenons as deduced from vocalizations

JEAN-PIERRE GAUTIER

Since the pioneering work of Struhsaker (1970) on cercopithecines, a number of primate studies have used vocalizations as phylogenetic criteria. They have mostly dealt with forest dwelling primates such as colobines (e.g. Wilson & Wilson, 1975; Struhsaker, 1981a; Oates & Trocco, 1983) and gibbons (Marshall & Marshall, 1976; Brockelman & Gittins 1984; Creel & Preuschoft, 1984; Creel *et al.*, 1984; Haimoff *et al.*, 1984; Marshall, Sugardjito & Markaya, 1984), while one study involved a comparison between langurs and forest macaques (Herzog & Hohman, 1984) and finally with South American tamarins (Hodun *et al.*, 1981; Snowdon *et al.*, 1986). This is indicative of the supremacy of vocal communication in forest environments, previously emphasized by Waser (1982a).

The use of behavioural traits as taxonomic characters relies primarily on their degree of genetic determination, and Kummer (1970) suggested that the 'best candidate for taxonomic usefulness is the so-called fixed action patterns . . .' such as vocalizations. Ontogenetic studies undertaken in both normal and socially deprived environments, along with neurophysiological studies (e.g. Jurgens, 1979) and structural analyses of hybrid calls (e.g. Gautier & Gautier, 1977; Bouchain, 1985), concur in giving a predominant role to genetic factors in determining the acoustic structure of calls (Newman & Symmes, 1982; Newman 1985). In contrast to the evidence available for many birds, vocal imitation in primates has not been documented. Long-term experiments using captive monkeys living in mixed-species groups (e.g. Bernstein, 1970 and personal observations), as well as the frequent occurrence of polyspecific troops among guenons in the wild (see Gautier-Hion, Chapter 23), show that, while these animals are able to

learn the significance of extraspecific signals, they do not change their own vocal patterns.

It has been pointed out for the vocalizations of birds and monkeys that different call types may have been subject to dissimilar kinds of selective pressures and thus have acquired different degrees of species-specificity (Marler, 1957, 1965; Struhsaker, 1970; Becker, 1982). The function of calls, the conditions in which calls are given, and the composition of the community to which species belong may all influence the patterns of selective pressure. The least species-specific calls can be expected to be those in which the emittor's identity is not important (e.g. alarm calls), or calls given at short distance which benefit from additional visual cues (e.g. aggressive calls). On the contrary, calls which serve to enhance inter-individual cohesion or which play a role in the reproductive isolation of a species can be expected to be the most species-specific. This explains why most studies of the phylogenetic implications of calls rely on the comparison of ritualized vocal patterns such as the long calls of tamarins, the songs of hylobatids or the loud calls of Cercopithecinae. In most African species, these highly stereotyped vocal behaviours mainly characterize adult males and require either an enlarged larynx, as in colobines (Hill & Booth, 1957), or extralaryngeal annexes, as in guenons (Gautier, 1971). However, if we suppose that such calls have evolved to prevent interspecific hybridization, they may be expected to diverge more among sympatric than among allopatric species, as has been shown by Marler (1957) for birds. From this perspective, adult male loud calls of guenons may be very good or very poor indicators of interspecific affinities, depending on the degree of sympatry among species. For example, relatively close similarity in the structure and temporal pattern of adult male loud calls is found in two species, *C. n. nictitans* and *C. mitis labiatus*, which are separated by 3000 km (Figure 12.1), while the sympatric *nictitans–cephus* or *mitis–ascanius* pairs of species show very close similarity in most of their repertoire, but adult male loud calls diverge markedly (see Figure 12.8 below).

To compare calls of different species raises a number of methodological problems. Firstly, when can we consider two calls to be structurally homologous? Is it at the point when we cannot distinguish them by ear or is it at the point when statistical analyses comparing a number of their acoustic parameters yield non-significant results? Measurements of such parameters can be made so numerous and precise that a statistical difference is almost inevitably found (see Gautier & Gautier-Hion, 1988), yet the distinctions may not be mean-

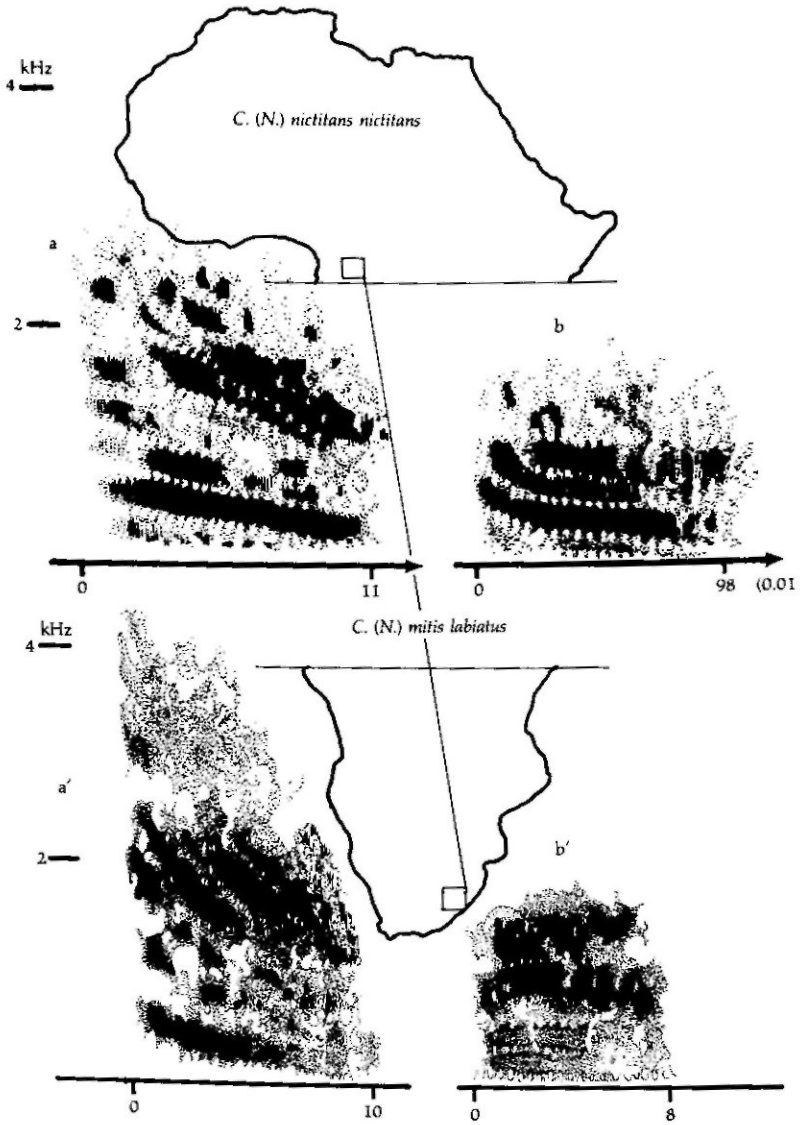


Figure 12.1. Spectrographic analyses of adult male loud calls in *C.n.nictitans* and *C.mitis labiatus* with the geographical locations of species (loud calls of first category: a, a'; loud calls of second category: b, b').

ingful with respect to monkeys as perceivers. Consequently, until more playback experiments are conducted to clarify the level of discrimination in monkeys, it seems wise to compare species on the basis of simple call characters.

Secondly, we have to decide whether two structurally and functionally homologous calls have the same phylogenetic implication as two structurally similar calls given in different functional contexts. In other words, does the function of calls give any additional information in determining interspecific affinities? In agreement with Marler (1957, 1965) and Struhsaker (1970), we believe this to be the case, and call function will be taken into account in the present study.

A third point concerns the problem of the frequency of use. For Kummer (1970), this cannot be used to infer phylogenetic relationships, since *a priori* social and environmental conditions may induce different rates of use of the same call. Thus, depending on population density, territorial vocal battles may or may not occur between *C. cephus* troops (personal observations). Another example is given by Hall (1965a), who thought that the silence of patas monkeys was an adaptive response learned by infants.

It has been frequently claimed that young calls are simply non-elaborated adult calls which cannot be used to deduce interspecific affinities. Indeed in most guenons, the vocal repertoire of immature animals differs notably from that of adults. In the latter, maturational processes change the structure of calls (especially by a decrease in pitch, Gautier & Gautier 1977). Changes in the frequency of use are also found and a number of calls given by immature animals are no longer given by adults, not because they are unable to give them (as can easily be shown in captive animals, personal observations), but because adults are no longer confronted with the usual context in which calls are uttered (Gautier & Gautier-Hion, 1982). Most adult male guenons only retain two or three call types, one of which (at least) has an original structure not derived from young calls.

In a social group, immature monkeys (which remain so for four to seven years) have their own role, and their repertoire of signals depends on and underlies these. Their calls therefore have to be considered as a full part of the species repertoire. The latter cannot be accurately understood by relying on adult calls only. In addition, studies of maturational changes in calls are very useful in determining homology between calls of different species. This is the case, for example, when two structurally different adult calls are shown to derive from the same kind of immature calls (Gautier, 1978).

We will see below that, in some cases, comparisons of adult and immature calls lead to opposite conclusions about interspecific affinities. If we hypothesize that adult calls are more derived than young ones, the latter could be considered as relatively poor indicators of phylogenetic affinities. We might also propose, however, that young calls and adult calls have been submitted to different selective pressures. In any case, resemblances in calls given by young of different species are sometimes so great that it seems difficult not to consider the calls as homologous, and to infer the existence of a common ancestor. Similar examples are given for insects by Hennig (1965), who points out that some phylogenetic affinities can only be shown by comparing immature stages.

Thus, in this study we will search for interspecific affinities by taking into account a number of calls, and examining their structure and temporal pattern, their overall functional context, their presence/absence in different age–sex classes as well as the behaviour which accompanies their emission, and morphological structures which allow it (see Martin, 1978).

Species studied

Twenty-three of the 28 species and about 40% of the 85 forms retained in the revised taxonomic list of Lernould (Chapter 4) are considered in this analysis. For 17 species, I will refer to personal records and observations and to other studies when available (see details in Table 12.1). For others, I will simply refer to the literature. However, great differences exist in the extent to which each species' repertoire has been studied. Data are very limited for about one third of species, and completely lacking for four guenon species: *C. denti*, *C. sclateri*, *C. dryas* and *C. salongo*.

Calls studied

Calls considered in this analysis belong to three functional systems (see the review in Gautier & Gautier, 1977). The first includes calls, given in pacific exchanges between members of the social unit, which serve in intragroup cohesion and pacific inter-individual relationships. The second consists of calls given in warning situations, which tend to spread among troop members through phonoresponses. The third category includes loud calls of the first category which are

Table 12.1. *List of species whose vocal repertoires have been studied*

Genus	Superspecies	Species	Subspecies	L[a]	References
Cercopithecus	*aethiops*	*pygerythrus*	*johnstoni*	3	19, 20
Cercopithecus	*aethiops*	*sabaeus*		3	4
Cercopithecus	*aethiops*	*tantalus*		2	16
Cercopithecus	*cephus*	*ascanius*	*ascanius*	1	11
Cercopithecus	*cephus*	*ascanius*	*katangae*	2	11
Cercopithecus	*cephus*	*ascanius*	*schmidtii*	2	17
Cercopithecus	*cephus*	*ascanius*	*whitesidei*	2	11, 13
Cercopithecus	*cephus*	*cephus*	*cephus*	3	9, 10, 12, 21
Cercopithecus	*cephus*	*cephus*	*cephodes*	1	11
Cercopithecus	*cephus*	*erythrogaster*		1	18
Cercopithecus	*cephus*	*erythrotis*	*camerounensis*	1	11, 21
Cercopithecus	*cephus*	*petaurista*	*petaurista*	2	6, 21
Cercopithecus	*diana*	*diana*	*diana*	2	6
Cercopithecus	*diana*	*diana*	*roloway*	1	11
Cercopithecus	—	*hamlyni*		3	11
Cercopithecus	*lhoesti*	*lhoesti*		2	11
Cercopithecus	*lhoesti*	*preussi*		1	11, 21
Cercopithecus	*lhoesti*	*solatus*		2	11
Cercopithecus	*mona*	*campbelli*	*campbelli*	1	21
Cercopithecus	*mona*	*campbelli*	*lowei*	2	2, 3, 6
Cercopithecus	*mona*	*mona*	*mona*	2	11, 21
Cercopithecus	*mona*	*pogonias*	*nigripes*	3	9, 12
Cercopithecus	*mona*	*pogonias*	*pogonias*	2	21
Cercopithecus	*mona*	*wolfi*	*wolfi*	1	11
Cercopithecus	—	*neglectus*		3	9, 12, 14
Cercopithecus	—	*nigroviridis*		2	11
Cercopithecus	*nictitans*	*mitis*	*stuhlmanni*	2	1, 17
Cercopithecus	*nictitans*	*mitis*	*labiatus*	1	23
Cercopithecus	*nictitans*	*nictitans*	*martini*	1	21
Cercopithecus	*nictitans*	*nictitans*	*nictitans*	3	7, 9, 12
Cercopithecus	—	*patas*		1	11, 15, 22
Cercopithecus	—	*talapoin*	?	3	8, 9

[a]L1: limited data; L2: detailed analyses; L3: exhaustive analyses.

References: 1. Aldrich-Blake, 1970; 2. Bertrand & Gautier, unpublished data; 3. Bourlière *et al.*, 1970; 4. Galat, 1975; 5. Galat, 1983; 6. Galat & Gautier, unpublished data; 7. Gautier, 1969; 8. Gautier, 1974; 9. Gautier, 1975; 10. Gautier, 1978; 11. Gautier, unpublished data; 12. Gautier & Gautier, 1977; 13. Gautier & Gautier-Hion, 1982; 14. Gautier-Hion & Gautier, 1978; 15. Hall *et al.*, 1965; 16. Kavanagh, 1981, 1986; 17. Marler, 1973; 18. Oates, 1985; 19. Seyfarth & Cheney, 1980, 1982, 1984; 20. Struhsaker, 1967e; 21. Struhsaker, 1970; 22. Struhsaker & Gartlan, 1970; 23. Gautier & Girolami, unpublished data.

only given by fully adult males in more or less ritualized sequences and appear to play a role in rallying troop members and spacing out troops within the population and the community.

Calls given in agonistic contexts have been excluded because they are poorly species-specific. Aggressive loud calls proper to adult males (loud calls of the second category, most often a bark type), which have been shown to derive from warning and aggressive calls of adult females and immatures (Gautier, 1974, 1978), have not been considered in detail, despite the fact that their temporal patterns and their associations with loud calls of the first category may be highly organized. Unfortunately, precise studies of such sequences could only be conducted for a few species to date.

Parameters studied

Highly species-specific calls such as adult male loud calls are easily distinguished on the basis of their overall structure. For other call types, whose interspecific similarities are greater, we have relied upon a few simple and stable parameters. These include: presence/absence of the call type; its adherence either to a low- or high-pitched register; the complexity of its acoustic structure (noisy components vs. pure tone, quavering vs. continuous calls); and temporal pattern of emission (unitary vs. multi-unitary calls or calls in series).

Methods of analysis

Our first approach was qualitative and phenetic. The distribution of calls or call characteristics among species permitted hierarchical clustering on the basis of overall similarity. Phylogenetic trends were then inferred from the discrimination between more or less derived forms depending on the vocal characteristics they share with species belonging to 'outgroups' such as Papioninae and Colobinae. In order to rationalize these preliminary inferences, we further applied a cladistic method involving two steps. First, the binary coding of calls and call characteristics allowed us to build a matrix of species by characters (Sneath & Sokal, 1973; Table 12.2). The data were then fed into several computer programs in the 'phylogeny inference package' (Felsenstein, 1982, 1985b).

Having tested a number of these programs, we have retained the Wagner parsimony method because it allows reciprocal changes between ancestral and derived states, as well as no statement about ancestral characters when data do not authorize any statement. Options also allow each character to be weighted differently. Given

Table 12.2. *Matrix of data. Binary coding of the 22 calls characters/15 taxa. Species with the same set of characters are grouped on the same row*

Type of call	Cohesion-contact calls								Adult males' loud calls 1st Category								High-pitched warning calls					
	Type 2			Type 6			2 + 6															
Type of character	quavered	non-quavered	+ noisy components	present	quavered	non-quavered	associated	merged	present	'boom'	'pyow'	'hack'	'gobble'	'whoop-roar'	'oop-uuh'	'hack-roar'	present	unitary chirp	multi-unitary chirp	twitters	whistle	pseudo-chirp
Number of characters	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Cercopithecus nigroviridis	1	0	0	1	1	0	0	0	1	0	0	0	1	0	0	0	1	1	0	0	0	0
C. nictitans/mitis	1	0	0	1	1	0	1	0	1	1	1	0	0	0	0	0	1	1	0	0	0	0
C. cephus/ascanius/erythrotis	1	0	0	1	1	0	1	0	1	0	0	1	0	0	0	0	1	0	1	1	0	0
C. petaurista	1	0	0	1	1	0	1	0	1	0	0	1	0	0	0	1	1	0	1	1	1	0
C. erythrogaster	1	0	0	1	1	0	1	0	1	0	0	1	0	0	0	1	1	0	1	1	?	0
C. diana	1	0	0	1	1	1	0	0	1	0	0	0	0	1	0	0	1	0	0	0	1	0
C. hamlyni	1	1	0	1	0	1	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
C. neglectus	0	1	0	1	0	1	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0
C. mona/campbelli	0	1	0	1	0	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
C. pogonias/wolfi	0	1	0	1	0	1	1	1	1	1	0	0	0	0	0	0	1	0	1	0	0	0
C. (aethiops)	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1
C. patas	0	1	0	1	0	1	1	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
C. talapoin	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
C. lhoesti/'solatus'	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0
C. preussi	1	0	?	0	0	0	0	0	1	0	0	0	0	0	1	0	1	1	0	0	0	0

these two possibilities, the program proposes several more or less parsimonious dendrograms which question the author and lead him to consider additional criteria to obtain the most relevant tree of interspecific affinities.

Characterization of calls

Low-pitched cohesion calls

This call system exists in all species and includes a more or less graded series of vocalizations which are only given in full by young animals (Figure 12.2). Within this series, type 2 calls (Gautier, 1975) are the most commonly uttered; they are mostly given without the support of visual clues and enhance intra-troop cohesion. Their frequency of use varies according to species, group size, age class, habitat structure and other ecological factors. Thus the silence of patas (Hall, 1965) as well as that of *C. neglectus* (Gautier & Gautier, 1977) and *C. preussi* (personal observations) may be viewed as adaptive strategies against terrestrial predators, while the frequent use of such calls by talapoins is attributable both to the great size of the troop and to the high structural density of the milieu they colonize, which impedes visual contact.

Type 2 cohesion calls can develop into type 3 lost calls given by young animals, when they lose contact with their mothers or with the social unit.

Gross examination of the structure of these calls allows the separation of species into two categories (Table 12.2). The first includes species which give discontinuous or quavered calls such as *C. nictitans* and *C. cephus* (Figure 12.2); the second includes species whose calls are continuous and non-quavered, such as *C. pogonias*, *C. neglectus* and *C. talapoin*, the latter species having the highest pitch (Figure 12.2). In *C. hamlyni*, both quavered and non-quavered calls are heard, the latter being given by infants, the former by adults (Figure 12.3).

Calls of *C. lhoesti* and *C. solatus*, which share a similar timbre with calls of *C. talapoin* due to the frequent association of high-pitched noisy components with the low register (Figure 12.4), are more difficult to include in one or other category defined above. However, detailed analysis reveals a quavering not easily discriminated by ear because of its very high rhythm and low amplitude (Figure 12.5).

High-pitched contact calls

A high-pitched register is found in all species; it groups a graded series of calls including those, like contact calls, which possess

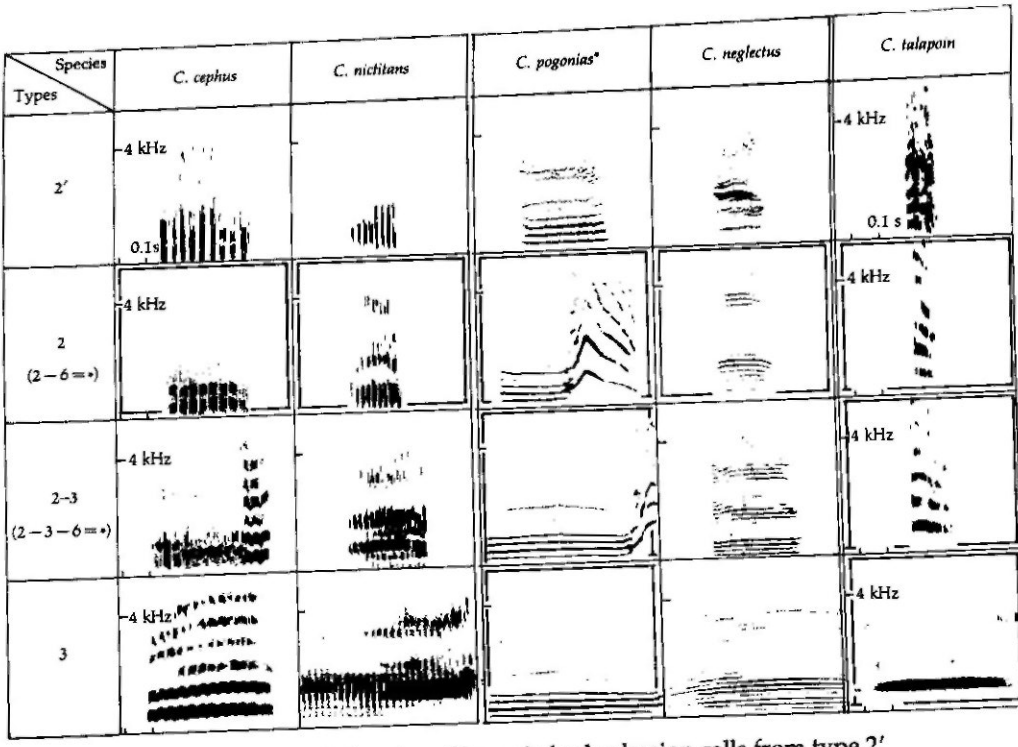


Figure 12.2. Graded series of low-pitched cohesion calls from type 2′ (young crouching calls), type 2 (usual cohesion calls) to type 3 (young lost calls). Calls are quavered in *C. cephus* and *C. nictitans*, and non-quavered in *C. pogonias*, *C. neglectus* and *C. talapoin* (from Gautier, 1975). Notice the pitch of talapoin calls which clearly differentiates the species while the structural similarities reveal two species pairs (*nictitans–cephus* and *pogonias–neglectus*). * indicates that type 2 and 6 are given in a single continuous call.

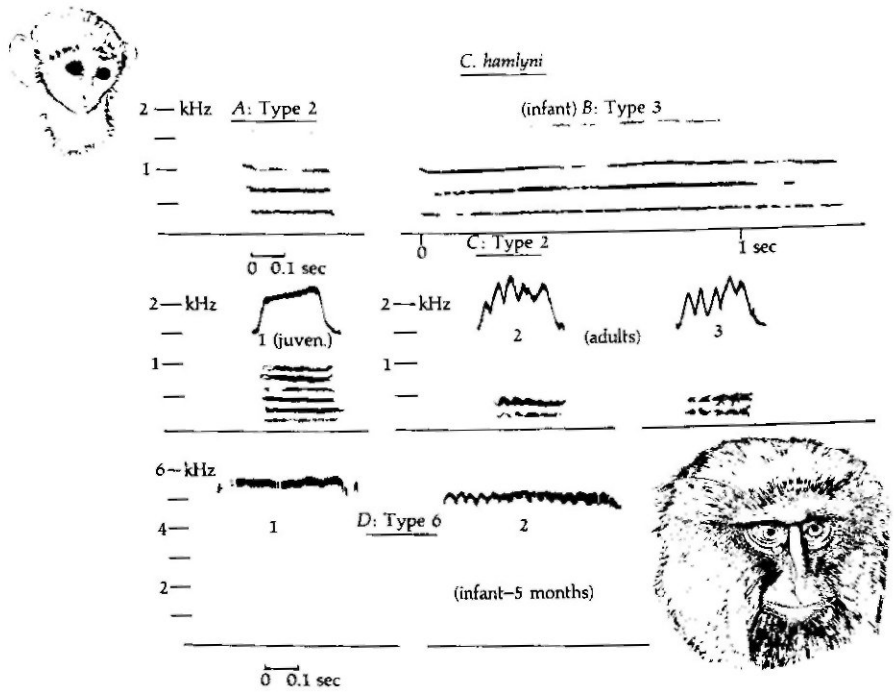


Figure 12.3. Ontogenetic changes in the structure of cohesion-contact calls (type 2 and 6) in *C. hamlyni*. *A* and *C*1: non-quavered type 2 calls of infants and juveniles; *B*: non-quavered type 3 call of infants; *C*2 and *C*3: quavered type 2 calls of adults (the quavering is clearly apparent on the amplitude modulation spectra); *D*1 and *D*2: non-quavered and quavered type 6 calls of a 5-month-old infant.

a well-defined acoustic structure, and other noisy and unstructured whistles and screams which are given in stressful situations and are poorly species-specific.

Together with the low-pitched calls, the high-pitched contact calls (type 6, Gautier, 1975) regulate inter-individual distances, especially between young and mothers. Their occurrence progressively decreases with age and they are not given by adult males of any species. In some species, they remain commonly used by subadults and adult females (e.g. members of the *mona* superspecies and *C. diana*, and to a lesser degree members of the *nictitans* and *cephus* superspecies and *C. nigroviridis*). In others, (e.g. *C. neglectus, hamlyni, patas*) they are only heard in infants. Moreover some species have not evolved these high-pitched contact calls, as in the case of the *aethiops* and *lhoesti* groups and of *C. talapoin*.

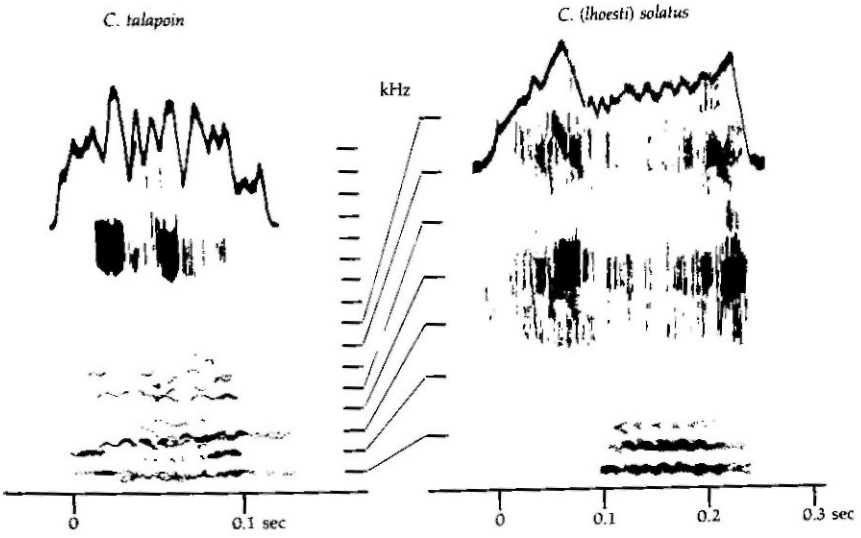


Figure 12.4. Similarity in calls structure of *C. talapoin* and *C. solatus* (here one unit of an agonistic call) resulting from the association of high-pitched noisy components (wide band filter) and low-pitched tonal quavered ones (narrow band filter). Calls of talapoins are 1.5 octave higher than those of *C. solatus*.

Thus if we consider the whole high-pitched register within the guenon group, there is a gradient from species in which only non-structured whistles and screams are heard, to those in which the whole system is given only by infants, to species in which all age classes can perform the whole register.

On the basis of the acoustic structure of their high-pitched contact calls, species can be divided into two clusters (Table 12.2) according to whether their calls are quavered or not (Figure 12.6). In *C. diana*, adult females and immatures are able to give both quavered and unquavered calls (Figure 12.7).

The whole cohesion-contact call system

Within species which possess both type 2 and type 6 calls, another distinction can be made as to whether or not they use the two calls in the same temporal sequences. For example, *C. nigroviridis* and *C. diana* mostly use the high-pitched type 6 call without association with type 2. All other species associate the two call types to a greater or lesser degree; the finest level of association is attained in the *mona* forms which merge the high and low-pitched components into a single continuous vocalization (see Figure 12.2).

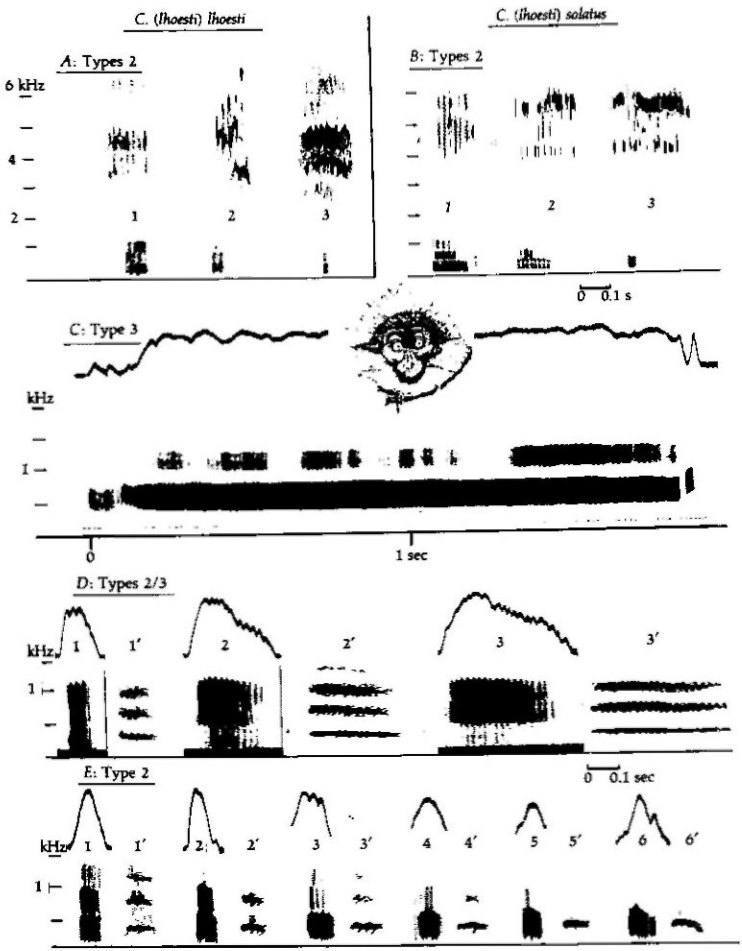


Figure 12.5. Cohesion call system of *C. lhoesti* and *C. solatus*. *A* and *B* 1, 2, 3 show the similarity between type 2 calls of the two species, which associate high-pitched noisy components and low-pitched tonal ones. *C*, *D* and *E*: detailed analyses of the low-pitched register of cohesion calls in *C. solatus* showing the variability in the duration and energy distribution of calls and the peculiar quavering characterized by a high rhythm and a weak amplitude (D1′–3′; E1′–6′: narrow band filter).

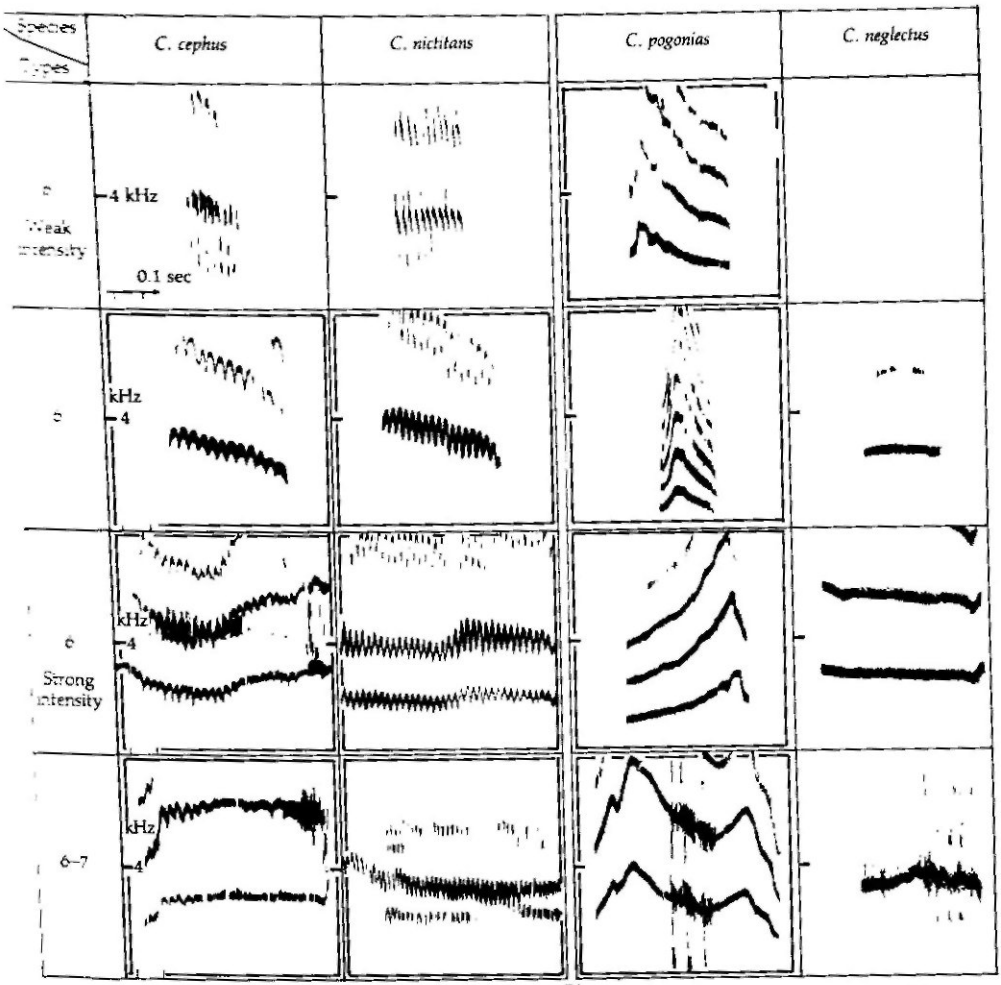


Figure 12.6. Graded series of high-pitched contact calls from type 6' (weakly voiced sounds), type 6 (usual contact call) to type 6–7 (call evolving towards distress vocalization). Calls are quavered in *C. cephus* and *C. nictitans* and non-quavered in *C. pogonias* and *C. neglectus*. Compare the amplitude of modulation of *C. pogonias* and *C. neglectus*' calls.

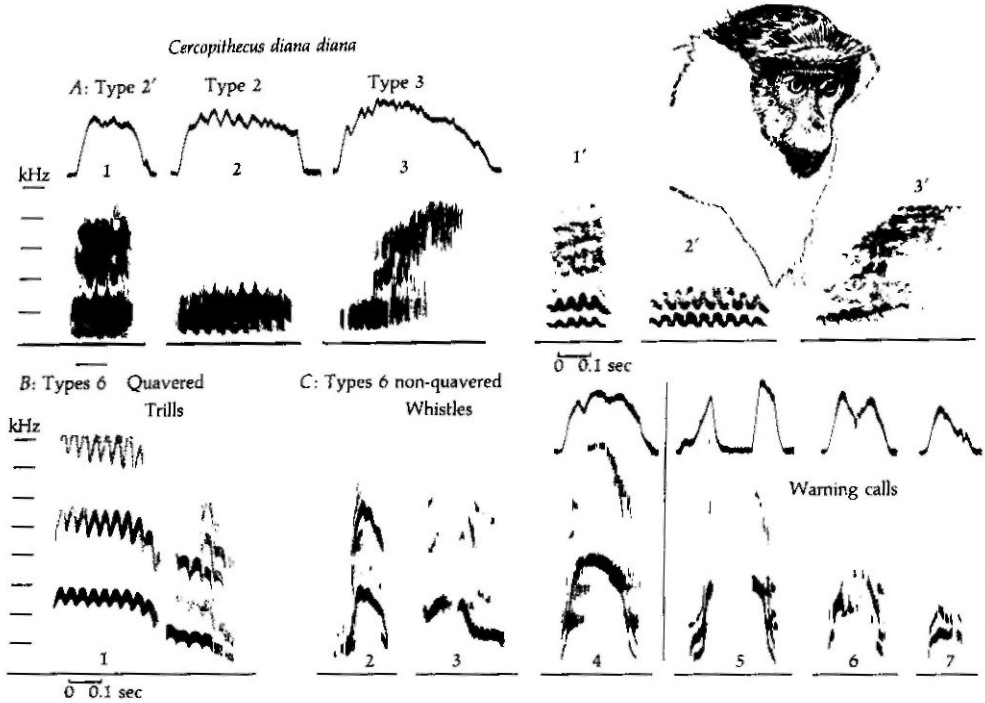


Figure 12.7. Quavered and non-quavered calls of *C. diana*. *A*: low-pitched quavered cohesion calls in young; (1, 2, 3 and 1′, 2′, 3′, respectively wide and narrow band filters) *B*: high-pitched quavered contact call; C2, 3, 4: whistling contact calls (given by a highly excited monkey); C5, 6, 7: warning calls.

Warning calls

Warning calls are found both in the low-, the medium-, and the high-pitched register. Only the latter will be considered here. The so-called chirps are the most frequent high-pitched alarms. They can be given either in single units (as in *C. nigroviridis*), in multi-units (as in *C. cephus*), or in series (twitters as in *C. petaurista*). A 'pseudo-chirp' which differs in its structure is also given by the *aethiops* forms. *C. diana* does not emit chirps in alarm situations but a very specific whistle. A similar whistling note ends the multi-unit squeak given, together with chirps, by young *petaurista* (Table 12.2). *C. hamlyni*, *C. neglectus*, *C. mona* and *C. campbelli* do not utter such high-pitched warning calls.

Loud calls

Loud calls of the first category (Gautier & Gautier, 1977) are present in all species except the talapoin, the *aethiops* forms, *C. lhoesti* and *solatus*. There is no relevant information for *C. patas*. According to J. S. Gartlan (personal communication), this species possesses a loud call very close to the 'oop-uuh' of *C. preussi* (Struhsaker, 1970); however, the 'whoo-wheer' described by Hall, Boelkins & Goswell (1965) and J. Chism (personal communication) does not possess the characteristics of loud calls and I have never succeeded in eliciting such calls in a captive group.

Eight species of the ten which possess loud calls utter 'booms', including the *mona* forms, *C. hamlyni*, *C. neglectus*, *C. nictitans* and *C. mitis* (Figure 12.8). However, the 'booms' of the two latter species are not really loud; they possess a lower pitch than in other 'boom' emittors and are not given in stereotyped sequences; moreover their function remains obscure. They have nevertheless been considered in the search for phylogenetic affinities because the production of such calls requires the development of large vocal annexes and a very complex behaviour implying a phase of inflating sacs which precedes the emission of the calls (Figure 12.9). Thus, it seems improbable that such a complex phenomenon has evolved independently in several taxa.

Loud calls of species which do not utter 'booms' are clearly distinct from each other (Figure 12.8). Within the same superspecies, however, species distinctiveness is often unclear. This is the case for the 'pyows' of the *nictitans* superspecies (Gautier, 1969; Marler, 1973). Within the *cephus* group, while it is difficult to distinguish the unitary calls of *C. cephus* ('mpaks', Gautier, 1975) and *C. ascanius* ('hacks', Struhsaker, 1970; Marler, 1973), the complex multi-unit hacks given by *C. petaurista* (personal observations) and *erythrogaster* (Oates, 1985) are clearly recognizable.

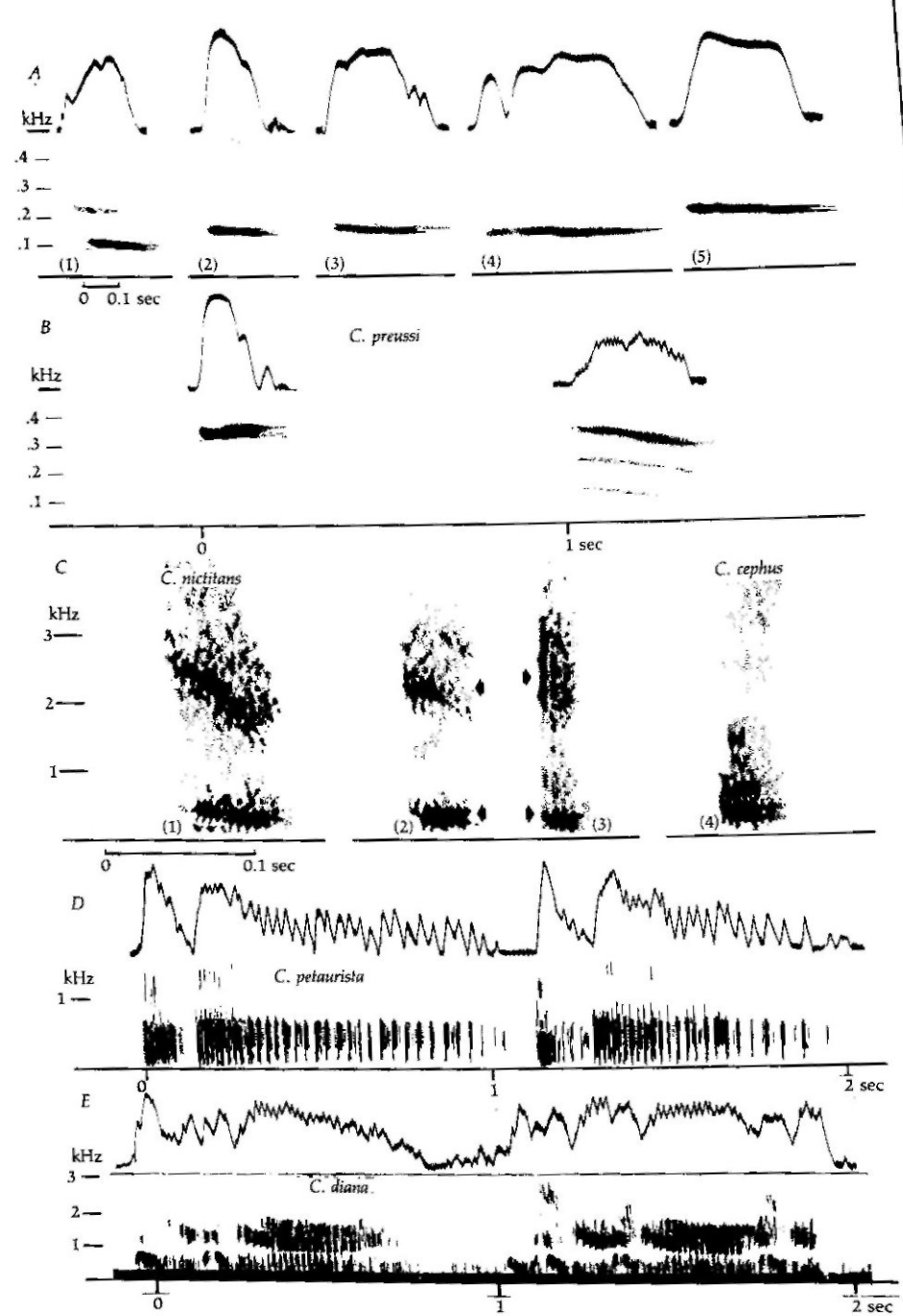


Figure 12.8. Loud calls of adult males. *A*: 'boom' of C. *nictitans* (1), C. *pogonias* (2), C. *campbelli* (3), C. *neglectus* (4), C. *hamlyni* (5); *B*: two-phase 'oop-uuh' of C. *preussi*; C: 'pyow' of C. *nictitans* (1) and 'hack' of C. *cephus* (4); transitional sounds given by *nictitans* (2) and *cephus* (3) which reveal interspecific affinities between the two species. *D*: 'hack roar' of C. *petaurista*; *E*: 'whoop roar' of C. *diana*.

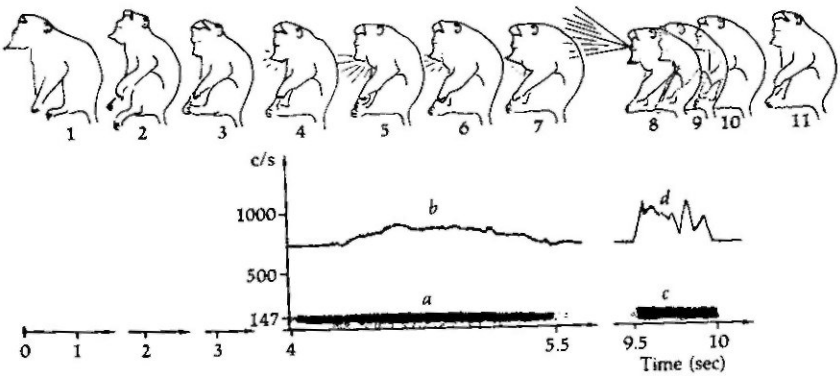


Figure 12.9. Behavioural sequence accompanying the uttering of 'boom' in *C. neglectus*. 1–3: preparatory phases; 4–7: inflation of the vocal sac accompanied by a sound (spectra *a* and *b*); 8–10: uttering of 'boom' (spectra *c* and *d*) and deflation of the vocal sac (from Gautier, 1971).

Data analysis

Before analysing the matrix (Table 12.2), we have to propose for each character the trait which can be considered as the ancestral state. We may assume that the presence of loud calls is a primitive trait because such calls are found in Asian and African colobines, mangabeys, baboons and even some macaques (see a review in Gautier & Gautier, 1977). The 'gobble' given by *C. nigroviridis* is quite probably a primitive trait too, since a similar call structure (given with or without a 'whoop') has been described in langurs (*Presbytis*: Sugiyama, Yoshiba & Parthasarathy, 1965; Ripley, 1967; Poirier, 1968; Horwich, 1976), in mangabeys (Chalmers, 1968; Gautier & Gautier, 1977; Quris, 1980; Waser, 1982a), in one baboon at least ('roar grunt' of *Papio papio*: Byrne, 1981) and in one macaque (*Macaca silenus*: Herzog & Hohman, 1984; Figure 12.10). Loud calls given by all the other guenons show a quite original structure and must be considered as derived traits.

The problem is not so clear for the cohesion-contact call system. In baboons, inter-individual relationships are sustained mainly by low-pitched grunts (Hall & DeVore, 1965; Byrne, 1981). Byrne distinguished four categories of grunts of which at least two (the two-phase grunt and the unitary grunt) are also found in drills and mandrills (Sabater Pi, 1972; Hoshino *et al.*, 1984). Hoshino *et al.* referred to a 'crow' in adult female mandrills, but we cannot decide whether or not this 'crow' is equivalent to the quavered 'crow' described in guenons.

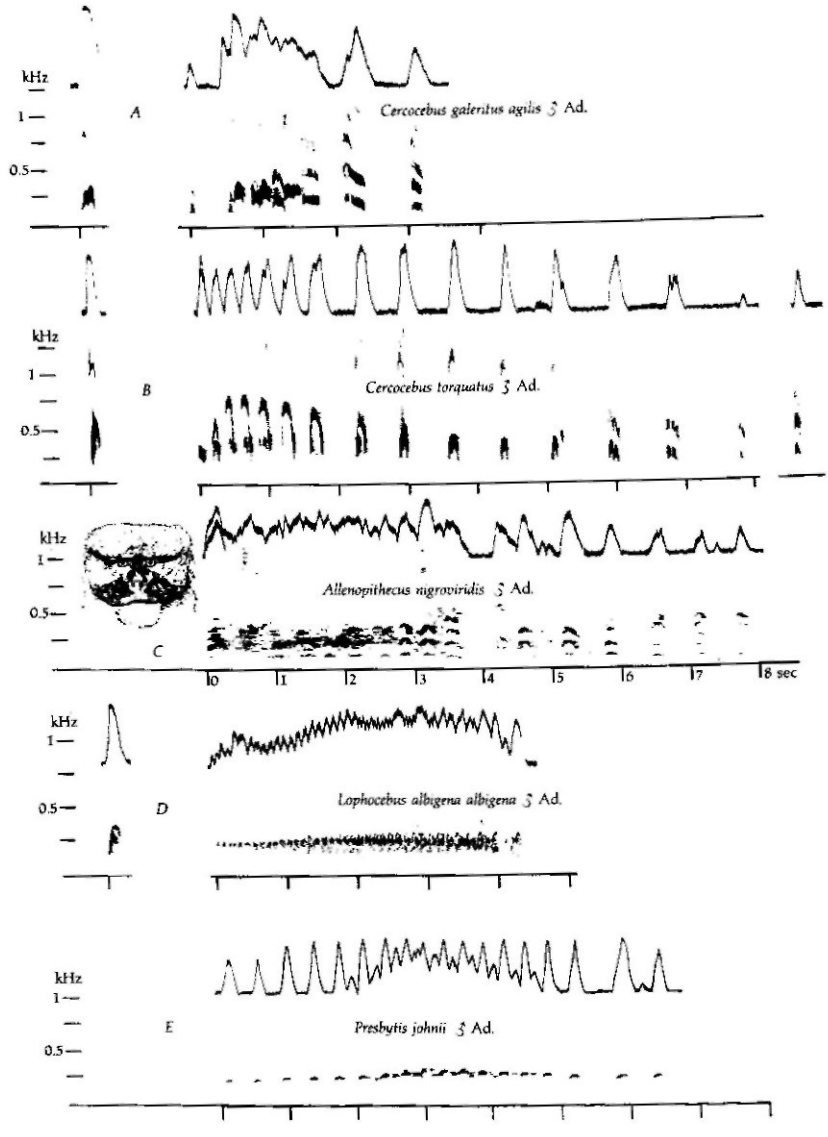


Figure 12.10. 'Gobble' calls in *C. galeritus* (*A*), *C. torquatus* (*B*), *C. nigroviridis* (C), *L. albigena* (*D*) and *P. johnii* (*E*). In *A*, *B* and *D*, the 'gobble' is preceded by an advertising 'whoop'.

In both structure and pattern of emission (series and choruses) the grunts of baboons are very close to those of mangabeys (Chalmers, 1968; Quris, 1973; Waser, 1975, 1976; Homewood, 1976; Gautier & Gautier, 1977; Deputte, 1986). The latter are clearly non-quavered. In mangabeys, as in baboons, the high-pitched register is not used in inter-individual relationships. Furthermore, in young baboons there are low-pitched continuous calls ('moans': Andrew, 1963; Ransom, 1976; Byrne, 1981; 'ooer': Hall & Devore, 1965) which are also found in young geladas (Richman, 1976); such 'moans' strongly resemble the non-quavered type 3 calls of *C. patas* and *C. neglectus* (Figure 12.11), *C. mona* forms and *C. hamlyni* (Figure 12.3).

In macaques, inter-individual relationships are regulated by a highly graded system of 'coo' sounds which are non-quavered and belong to a medium-pitched register (e.g. *Macaca mulatta, fuscata, nemestrina, speciosa* and *silenus*: Rowell & Hinde, 1962; Grimm, 1967; Bertrand, 1969; Green, 1975; Hohmann & Herzog, 1985). In these species the high-pitched register is rarely used and poorly structured. Some exceptions do exist, as in the case of the 'food call' of *M. sinica* (Dittus, 1984) which associates low- and high-pitched components as do *mona* forms.

African colobines use well developed high-pitched calls (Marler, 1970, 1972; Struhsaker, 1975, 1981a). In most cases their function remains poorly understood. In *Colobus guereza*, however, they are used in social interactions (Oates & Trocco, 1983). Such calls are highly modulated but do not appear to be systematically quavered. Finally high-pitched quavered calls are uttered by many New World monkeys (see a review in Oppenheimer, 1977).

Consequently, two hypotheses have been envisaged for the cohesion-contact system (Table 12.3). In the first, the presence of a high-pitched register and the presence of non-quavered calls have been considered to be primitive traits. This is also the case for the fusion of continuous calls of type 2 and type 6 found in the *mona* group, since a very similar call has been described in *Macaca sinica* (Dittus, 1984). In the second, we have left the problem unsolved by putting in question marks. In both cases, the presence of loud calls and the presence of a 'gobble' call have been considered as the ancestral state.

The two hypotheses have been tested with and without taking warning calls into account. The latter has been done in order to avoid establishing biased affinities which could result from patterns of convergence (Table 12.3). In the first hypothesis, high-pitched warning

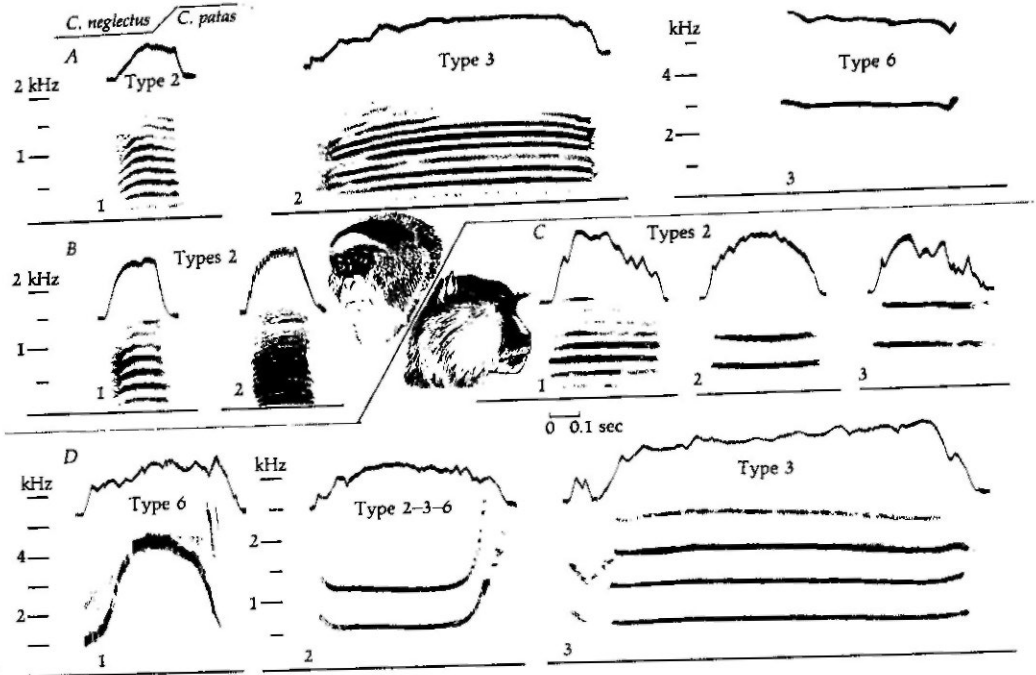


Figure 12.11. Cohesion-contact call system in *C. neglectus*, A, B, and *C. patas*, C, D. In both species, calls are non-quavered and look very similar. However the higher pitch of patas calls give them a timbre quite different from those of *C. neglectus*; in both species, the pitch of Type 2 calls decreases with aging (A_1, C_3: infants; B_1, C_2: juveniles; B_2, C_1: adults) (with the permission of J. Kingdon for the drawings of the heads).

Table 12.3. *Ancestral states of the different characters*

Characters	Ancestral state of characters			
	hypothesis 1		hypothesis 2	
	− WC	+ WC	− WC	+ WC
Loud call	P	P	P	P
'Gobble' call	P	P	P	P
High-pitched type 6 call	P	P	?(P)	?(P)
Quavering of type 2 and 6 calls	A	A	?(P)	?(P)
Highly-pitched noisy components (type 2)	P	P	?(A)	?(A)
Type 2 & 6 association	A	A	?(A)	?(A)
Type 2 & 6 fusion	P	P	?(A)	?(A)
High-pitched warning calls	/	P	/	?(P)
Unitary warning calls	/	P	/	?(P)
Other types of warning calls	/	A	/	?(A)
Number of characters	16	22	16	22
Number of steps	35	44	29	39

(): ancestral state proposed by the program; P: plesiomorphic; A: apomorphic. WC: warning calls. (See text.)

calls have been considered as primitive, except for those given in multi-units or in series; in the second, we did not make a decision.

Finally, we have chosen to give the same weight (i.e. 1) to all characters except the presence of a 'boom' call which is a highly specific feature of many guenon taxa (it has been weighted 3 or 4 respectively when warning calls are considered or not); and the 'gobble' call (weighted 3) which is the only loud call not specifically belonging to the guenon group.

Interspecific affinities

Two different phylogenetic trees are obtained from the four combinations derived from the two hypotheses and from the two sets of characters (Table 12.3). In fact, the same tree is produced from the second hypothesis when taking warning calls into account and when leaving them unweighted, and from the first hypothesis when warning calls are not considered (Figure 12.12).

Furthermore, the second hypothesis (in which we do not make a decision regarding the primitive nature of the cohesion call system or on the warning call system), leads to more parsimonious dendrograms (29 steps instead of 35 for the set of 16 characters and 39 steps instead of 44 for the set of 22 characters, Table 12.3). This second hypothesis

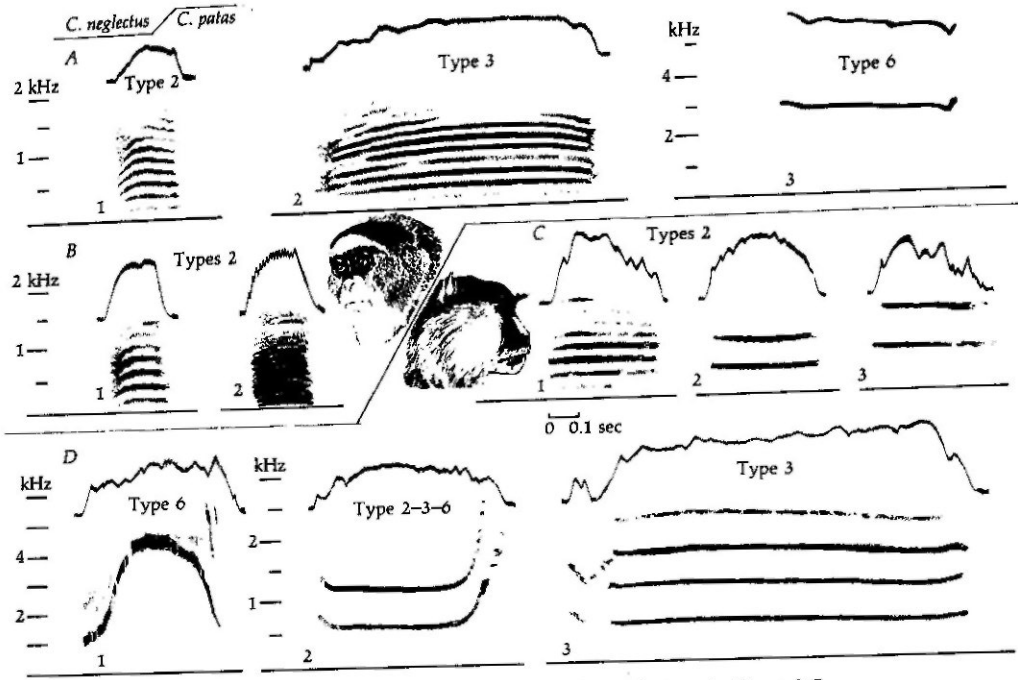


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Quavering of type 2 and 6 calls	A	A	?(P)	?(P)
Highly-pitched noisy components (type 2)	P	P	?(A)	?(A)
Type 2 & 6 association	A	A	?(A)	?(A)
Type 2 & 6 fusion	P	P	?(A)	?(A)
High-pitched warning calls	/	P	/	?(P)
Unitary warning calls	/	P	/	?(P)
Other types of warning calls	/	A	/	?(A)
Number of characters	16	22	16	22
Number of steps	35	44	29	39

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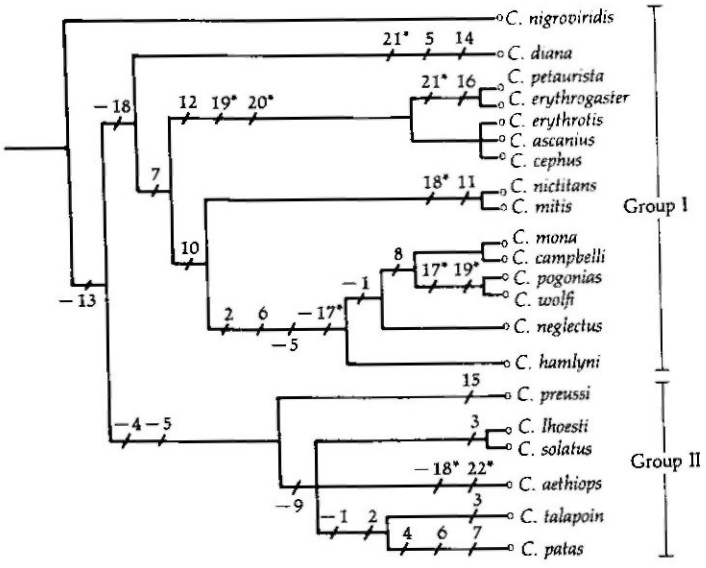


Figure 12.12. Dendrogram resulting from hypothesis 2 and hypothesis 1 (when warning calls are not considered, see text). Numbers refer to the different steps of characters listed in Table 12.2 *: characters referring to warning calls.

agrees with the polarity we proposed in the first hypothesis on two points: the ancestral state of high-pitched contact calls and of high-pitched unitary warning calls, and the derived trait of type 2 and type 6 association and of multi-unitary warning calls (Table 12.3). Conversely, it proposes that the quavering of calls is the ancestral state, suggesting that this character, which is found in New World monkeys, has been retained by guenons, while it has largely disappeared in colobines and papionines, possibly as a result of their larger body weight.

Let us consider first dendrogram 1 (Figure 12.12). *C. nigroviridis* appears as the closest to the ancestral form; this is mostly due to its 'gobble' call. Guenons are then separated into two groups according to whether they possess both loud calls and high-pitched contact calls (group 1, Figure 12.12) or not (group 2, Figure 12.12). The *diana* monkey is the first species to be separated from group 1. It is in fact well distinguished by its whistle alarm, its ability to give both quavered and unquavered contact calls and by its loud call.

The next step separates the *cephus* members from the 'boom' emitters. Within the *cephus* superspecies, two clusters of affinities are found: between *C. cephus*, *C. ascanius* and *C. erythrotis* on the one hand,

and between *C. petaurista* and *C. erythrogaster*, the two westernmost forms, on the other. Among the 'boom' emittors, the two closely related *nictitans* and *mitis* species are easily distinguished from the other taxa. The latter indeed share many derived traits, especially the disappearance of the ability to utter quavered calls and high-pitched alarms. In the latter taxa, the greatest similarity is found within the *mona* superspecies, which is distinguished from *C. neglectus* and *C. hamlyni* by the acquisition of a more graded repertoire while retaining the same timbre for most calls (Gautier, 1975). Two related species pairs are found in the *mona* group, with *pogonias* and *wolfi* on the one hand (which have acquired, probably by convergence, the ability to give high-pitched binary alarms), and *mona* and *campbelli* on the other hand (which share noisy warning sneezes, not considered in the present analysis).

Most calls of *C. neglectus* and *C. hamlyni* are very similar in their overall structure. The only difference observed from the characters used results from the ability of subadult and adult *C. hamlyni* to give quavered type 2 calls which resemble those of *C. nictitans*, while non-quavered calls like those of *C. neglectus* are given by young animals (Figures 12.2, 12.3). This ability confers on *C. hamlyni* an intermediate position between the *nictitans–cephus* group and the *mona–neglectus* group.

Species of group 2 are grouped more by the loss of a number of characters such as high-pitched sounds and loud calls (except for *C. preussi*) than by the presence of shared traits. The most surprising result concerns the relative position of the three members of the *lhoesti* superspecies, which are not grouped together. In fact, detailed observations and recordings made on captive groups of *C. lhoesti* and *C. solatus*, as well as field observations on the latter species (unpublished data), have shown that their repertoires are very similar and it is highly improbable that adult males of these species are able to utter loud calls of the first category as *C. preussi* does.

The repertoire of *C. preussi* is poorly known. Struhsaker (1970) did no more than mention the alarm call and the adult male loud call. After one hundred hours spent in contact with wild troops, we recorded, in addition to these two calls, many unitary barks given by adult males and only 14 cohesion calls given by females and immatures. Loud calls of the first category were found to be quite frequent; they were heard every evening and, on such occasions, up to ten adult males could respond to each other (Galat, Galat-Luong & Gautier, unpublished data). Thus the ability to give loud calls as well as the timbre of their

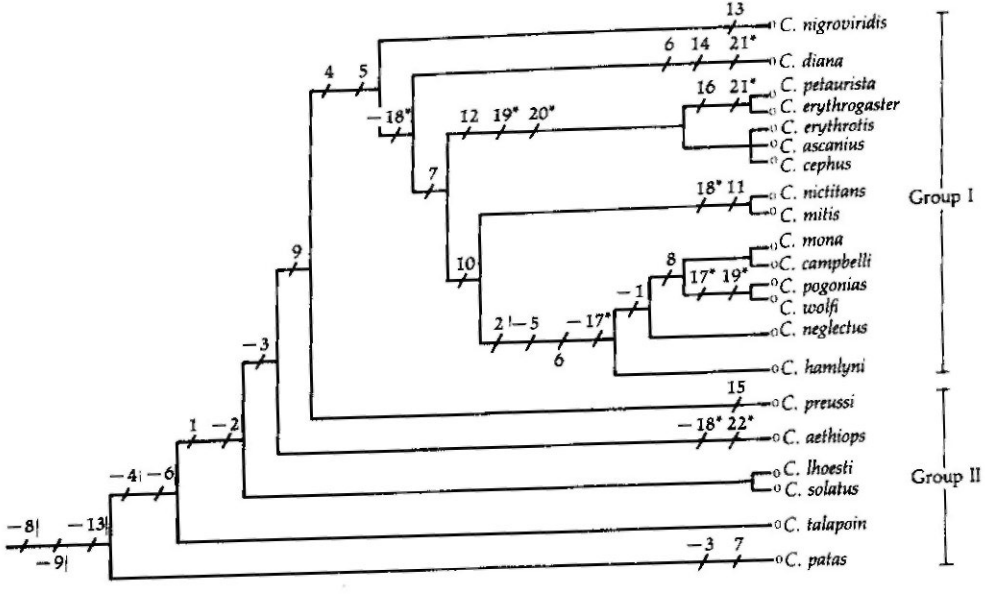


Figure 12.13. Dendrogram resulting from hypothesis 1 when warning calls are considered (see text; legends as in Fig. 12).

cohesion calls, which more closely resemble calls of *nictitans* than of *lhoesti* and *solatus*, makes the repertoire of *C. preussi* clearly different from those of the two other related forms.

The affinities among the other species of group 2 are less clear. In the preliminary treatment of our matrix we observed that a small change of a call character was able to modify the relative branching of these species, while that of the species of group 1 was regularly confirmed. Cohesion calls of *C. lhoesti*, *C. solatus* and *C. talapoin* possess high-pitched noisy components. On the other hand, *C. talapoin* and *C. patas* have in common the absence of quavering in their cohesion calls.

The second dendrogram is obtained under the first hypothesis when taking account of warning calls (Figure 12.13). In fact, the species of group 1, as well as their relative branching, remain the same as in the previous tree, except for *C. nigroviridis*. The latter species becomes the most ancestral form of this group instead of being the most ancestral form of all guenons. All species of group 1 share the ability to give loud calls and are differentiated from species of group 2 by the acquisition of high-pitched contact calls.

Another important difference is that in the second dendrogram, species of group 2 no longer appear as a group of species related to group 1 but as a paraphyletic group which shares primitive traits (synplesiomorphy, Hennig, 1965). In this group, the patas and the talapoin are closest to the ancestral form of guenons and *C. preussi* the closest form to species of group 1 due to its ability to give loud calls. In this group, *C. lhoesti* and *C. solatus* are the first species to acquire quavering in their calls.

Comparison with other phylogenies

Interspecific affinities among guenons obtained in the first dendrogram (Figure 12.12) are in close agreement with those deduced from karyological evolution (Dutrillaux *et al.*, 1982, Chapter 9). Both studies agree on:

- the most ancestral position of *C. nigroviridis*;
- the same two clusters of species, except that in the chromosomal study, *C. nigroviridis* is rooted at the base of group 1 and not at the base of the two groups;
- the primitive position of *C. diana* within group 1;
- the grouping of *C. neglectus* close to the *mona* forms;
- to some extent, the intermediate position of *C. hamlyni* between the *nictitans–cephus* group and the *mona–neglectus* group.

They mostly differ by:

the affinities of *C. hamlyni*, whose karyotype differs from that of *C. neglectus* by 14 chromosomal rearrangements and the possession of a greater number of chromosomes while the vocal repertoires of the two species are very similar;

the fact that the same karyotype has been observed for *C. lhoesti* and *C. preussi* while the karyotype of *C. solatus* differs by three chromosomal rearrangements of which one is shared with *C. patas*; conversely, the vocal repertoires group *lhoesti* and *solatus* and separate *preussi*;

the early departure given to *C. talapoin* in group 2 by the karyological study and to *C. preussi* in our dendrogram.

The results obtained in the second dendrogram (Figure 12.13) also agree with the findings of Dutrillaux *et al.* in that we found exactly the same two clusters of species, with *C. nigroviridis* more closely related to the species of group 1. However, in our study all forms of group 2 appear as more primitive than all guenons of group 1.

Comparing our results with Ruvolo's findings (Chapter 7) based on protein electrophoretic analysis, several points can be made. While Ruvolo also agrees that *C. neglectus* shares a common ancestor with the *mona* forms (the latter being clustered into the same two species pairs as in our study), she finds that *C. diana* is included in this large species group, being either less or more derived than *C. neglectus*, depending on the method of analysis used.

Furthermore, whereas we find close similarities between the repertoires of *C. hamlyni* and *C. neglectus*, none of the three analyses proposed by Ruvolo brings these species together. In addition, electrophoretic data always link *hamlyni* with *lhoesti* (probably *C. lh. lhoesti*).

Ruvolo points out that whatever the method of analysis used (phenetic or cladistic) the branching of *C. petaurista* remains uncertain, being alternately linked with *cephus* or with *albogularis* (a form of the *nictitans* superspecies) or having a primitive position with respect to all *nictitans* and *cephus* forms. Furthermore, *nictitans* and *cephus* are found either to have a common ancestor or to derive from one another, with either a *cephus* form or a *nictitans* form being the most derived. To some extent, we are confronted with the same problem (see below); our analysis also agrees that *C. petaurista* (together with *erythrogaster*) is distinct from *cephus*, *ascanius*, and *erythrotis*, a separation which is not found by karyological analysis.

From outgroup comparisons, Ruvolo concludes that *C. aethiops*, *C. patas* and *C. talapoin*, together with *C. nigroviridis* are 'equally primitive'

while the affinities she finds between these species differ according to the type of analysis used. However, in her two phenetic dendrograms, Ruvolo finds affinities between talapoin and patas as shown on our dendrogram 2 (Figure 12.13). Finally, it is worth noting that measurements of skull and teeth in guenons show affinities between *C. talapoin*, *C. patas*, and *C. lhoesti* (Martin & MacLarnon, Chapter 10).

Toward a consensus tree

Comparing our two dendrograms, a consensus tree can be proposed for phylogenetic affinities among guenons by considering additional vocal criteria not used in our matrix and some morphological and behavioural criteria (see Figure 12.15 below). In producing such a tree, we will particularly discuss the controversial points arising from the comparison of our results with those of the genetic studies of Ruvolo and Dutrillaux *et al.*

Our two dendrograms agree in clustering guenons into two groups (Figures 12.12, 12.13). Furthermore, the relative branching of the fourteen species of group 1 is exactly the same in the two dendrograms; only the position of *C. nigroviridis* differs.

The main differences rest on (1) the relative branching of the different species. In the first dendrogram, *C. nigroviridis* is the most primitive form of all guenons; in the second it is most ancestral to group 1, while all forms of group 2 are more ancestral than those of group 1; (2) the relationship between the two groups and within species of group 2, the latter being considered as a paraphyletic group in dendrogram 2.

Let us consider first the species of group 1. Within this cluster, the grouping of *C. diana* with the *neglectus–mona* cluster proposed by Ruvolo, which could be accepted on the basis of some similarities in coat patterns, is not supported by their vocal repertoires. Although *C. diana* calls are well differentiated from those of all other guenons, their closest affinities are found with the cohesion-contact calls of *nictitans* and *cephus* superspecies and especially with *C. petaurista*. It should be noted that both *petaurista* and *diana* give a whistle alarm call and associate a roar with their loud calls. The branching of *diana* in a relatively primitive position is not due to similarities with outgroups but to the fact that this monkey is able to give both quavered and non-quavered high-pitched sounds. A similar 'outlier' position is given to *C. diana* on the basis of skull and teeth measurements (Martin & MacLarnon, Chapter 10).

Similarly, we cannot propose a different grouping for the cluster including the four species of *mona*, *C. neglectus* and *C. hamlyni* and we

have no argument to link *hamlyni* with *lhoesti* as Ruvolo does. Must we attribute the close similarity between the repertoires of *hamlyni* and *neglectus* to patterns of convergence? Both species are semi-terrestrial forest monkeys, but the equivalence of their ecological niches remains unknown as C. *hamlyni* has never been studied in the field. The affinity between these two species is also supported by at least two other remarkable features: the very similar coat colour of their young which changes with age, and the existence of a very similar olfactory marking behaviour (see Loireau & Gautier-Hion, chapter 14). However, the two latter traits may also be invoked to link C. *hamlyni* and C. *neglectus* with species of group 2, since a distinct coat colour (although less differentiated than in C. *hamlyni* and C. *neglectus*) is also found in young C. *lhoesti*, whereas olfactory marking behaviour is also present in C. *aethiops* (and in C. *nigroviridis*). In addition, medium-pitched warning calls described in C. *patas* ('huh' calls: Hall *et al.*, 1965) and in C. *aethiops* (Struhsaker, 1967e) are also uttered by C. *hamlyni* (personal observations). Finally, we have seen that very similar 'moans' are given by C. *patas*, C. *hamlyni*, C. *neglectus* and C. *mona* (see Figure 12.11).

The relative branching of the C. *cephus* and C. *nictitans* superspecies which, in our analysis, results only from the presence of a 'boom' call in the latter species is worth mentioning. Detailed studies have shown that the repertoires of C. *mitis stuhlmanni* and C. *ascanius schmidti*, two sympatric species from East Africa (Marler, 1973), as well as of C. *n. nictitans* and C. *c. cephus*, their ecological equivalents in Gabon (Gautier, 1975, 1978), are very difficult to distinguish (see Gautier & Gautier-Hion, 1988). Thus it may be better to propose a common ancestor to the *nictitans* and *cephus* forms with the latter having lost the ability to utter 'booms' (Figure 12.15). This hypothesis is the more plausible as, contrary to many guenons, *cephus* monkeys have undergone a drastic reduction of their vocal annexes which, as demonstrated experimentally, operate in the uttering of 'boom' calls (Gautier, 1971).

The early departure of C. *nigroviridis* found by morphological characters (especially cranial measurements and the presence of a sexual skin: Verheyen, 1962; Hill, 1966; Martin & MacLarnon, Chapter 10) and by genetic studies, is supported in our dendrograms by the ability of this species to give a 'gobble' call. Other vocalizations confirm this, especially the copulation quaver given by females, which looks very much like that given by red colobus females (Struhsaker, 1975). Furthermore, it seems most probable that the swamp monkey is closest to species of group 1 as found in dendrogram 2 (Figure 12.13). Cohesion-contact calls and unitary warning chirps of this species are very similar

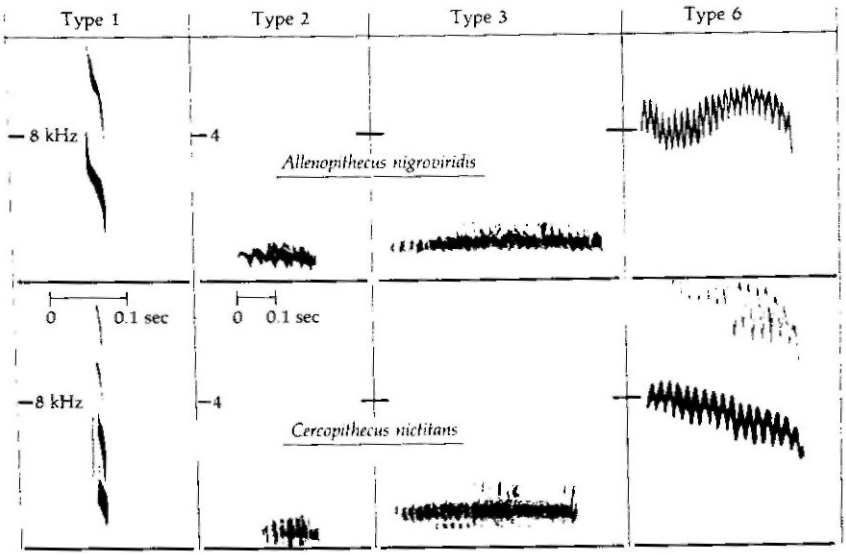


Figure 12.14. Structural similarities of warning calls (type 1), cohesion calls (type 2 and 3), and contact calls (type 6) given by *C.* (*A.*) *nigroviridis* and *C. nictitans*.

to those of *C. nictitans* (Figure 12.14) while no obvious resemblance is found between its repertoire and those of any other species of group 2. However, the metallic components of its unitary chirps give them a timbre similar to that of equivalent talapoin calls. In addition, the swamp monkey and the talapoin are the only guenons in which high-pitched chirps are retained in adult males. Both species also have a greenish coat, are the least brightly coloured of all guenons, and possess a sexual skin.

The primitive nature of *C. talapoin* found by the genetic and anatomical studies and in our dendrogram 2 is confirmed by a number of characters of its vocal repertoire. The latter shows astonishing similarities with those of macaques, while there is little analogy in the physical structure of talapoin calls and those of other forest guenons (Gautier, 1974). Thus, the 'clear calls' and the 'coos' of *M. mulatta* (Rowell & Hinde, 1962). the 'coos' and the 'high-pitched musical notes' of *M. nemestrina* (Grimm, 1967), the 'coos' of *M. speciosa* (Bertrand, 1969) and of *M. fuscata* (Green, 1975) as well as the 'whoo' call of *M. silenus* (Hohman & Herzog, 1985) resemble the type 2 cohesion calls of talapoins. All these calls possess a similar timbre and a fundamental tone around 1 kHz. Furthermore, in the talapoin as in some macaques, these calls may be given in choruses. The lost calls of young talapoins,

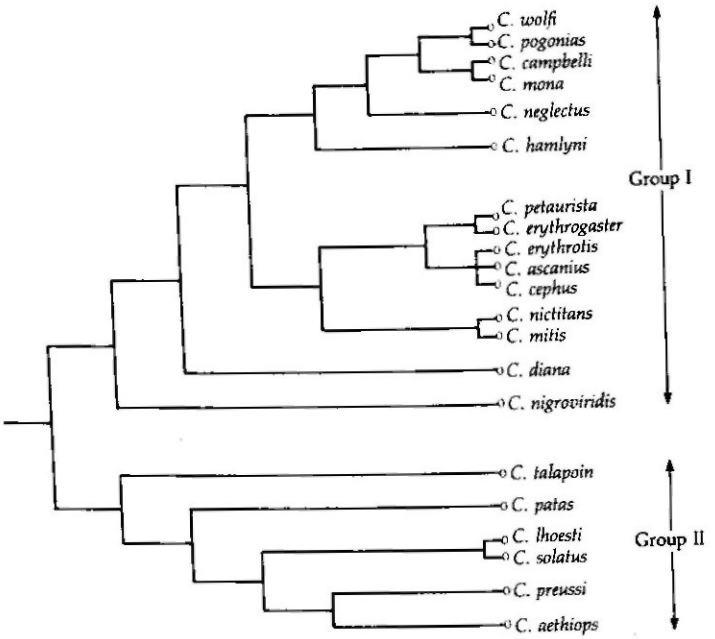


Figure 12.15. Consensus tree of phylogenetic affinities among guenons (see text).

young rhesus and young *nemestrina* all possess an identical structure and are difficult to distinguish. Finally, both in talapoin and rhesus, there is a graded call system arising in approach–avoid behaviours which is structurally similar, and the threat calls (type 4 in talapoin, and 'pant threat' in rhesus) have the same rhythm (4 to 5 units/sec) in both species.

These macaque-like calls confer on talapoin monkeys a position close to the common ancestor of papionines and cercopithecines. Given that we can make no statements about the relative position of talapoins and swamp monkeys, and that while the latter shows affinities with group 1 the former has little in common with those species, it seems justifiable to keep the two clusters of monkeys, with C. *nigroviridis* rooted at the base of group 1 and C. *talapoin* at the base of group 2 (Figure 12.15).

On an acoustic basis, there are also some similarities between C. *talapoin* and young C. *patas*, especially in their cohesion calls which are not quavered (compare Figures 12.2, 12.11). These two species, like C. *nigroviridis* (and also at least C. *solatus*) also give frequent copulatory calls. Such calls are known in colobines (Struhsaker, 1975), in baboons

(Hamilton & Arrowood, 1976), in mangabeys (Gautier & Gautier, 1977), and in macaques (Deputte & Goustard, 1980) and may thus be considered as an ancestral trait.

The early divergence of the patas monkey also relies on the fact that adult males give a two-phase bark which sounds similar to that of baboons (Hall & DeVore, 1965). Thus, data on vocal repertoire, concurring with other studies, suggest that both talapoin and patas are close to the ancestral form of guenons and, as found by Ruvolo in her phenetic analyses and by Martin & MacLarnon from skull and teeth measurements, share several primitive characteristics. Conversely, these two species have few vocal characters in common with the other species of group 2 (see Table 12.2).

We lack arguments to specify the relative position of the *lhoesti* and the *aethiops* superspecies, except that both taxa have a well developed use of the low-pitched register. Furthermore, there is some resemblance in the pattern of emission of cohesion calls given by *C. l. lhoesti* and *C. aethiops*. Finally, we currently have no data to link closely the three *lhoesti* forms: the affinity found between *lhoesti* and *solatus* in vocalizations clearly disagrees with the result of the karyological evidence which links *lhoesti* and *preussi*.

Conclusion

The consensus tree (Figure 12.15) based on characteristics of their vocal repertoires separates guenons into two groups of species which are the same as those found in the chromosomal evolution proposed by Dutrillaux *et al.* Notably, species of group 1 are all forest-dwelling monkeys and, except for three of them (*C. nigroviridis*, *C. neglectus* and *C. hamlyni*) fully arboreal species. Group 2 on the other hand includes the two savanna guenons, the three semi-terrestrial forms of *C. lhoesti* and the talapoin, a monkey which is mainly observed in the lower strata of the riparian forest and which shows several terrestrial traits in its locomotor behaviour (Rollinson, 1975).

All species of the latter group, together with *C. nigroviridis*, show ancestral traits in their vocal repertoires, but their relative affinities are still unclear. As in the study by Ruvolo, there is conflict in the placement of species which are relatively more primitive.

Recently, the repertoires of arboreal forest species have become better known and this knowledge has clearly influenced the choice of the criteria used in our analysis. Though these criteria are appropriate for arboreal forms, they may be less reliable for terrestrial and semi-terrestrial forms which are more primitive (see also Creel & Preuschoft,

1984). One cannot exclude, however, the possibility that these plesiomorphic species are not closely related to one another.

Nevertheless, our conclusions are strengthened by the good agreement among phylogenies drawn from studies in different disciplines. These conclusions also agree with the result of recent studies on fossils, forest history and paleoecology (see Leakey, Chapter 1; Hamilton, Chapter 2; Pickford & Senut, Chapter 3) which all indicate that early cercopithecoids quite likely evolved in semi-arid open woodlands and savannas and were semi-terrestrial forms.

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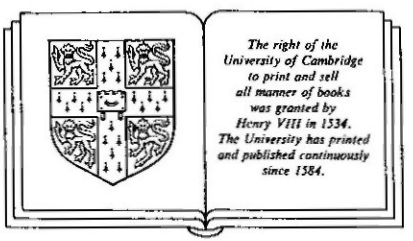
A primate radiation: evolutionary biology of the African guenons

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